18} Breast Examination

Goal:

At the end of this lecture, each student should acquire complete knowledge, skills and positive attitude toward breast examination and how to apply procedure breast examination.

Objectives:

- **At the end of this lecture, each student will be able to:**
  - Define of breast examination.
  - List purpose of breast examination.
  - Define indication of breast examination.
  - Discuss types of breast examination.
  - Discuss risk factors of breast illness
  - Determine of time for breast examination.
  - List equipment of breast examination.
  - Apply procedure of breast examination.

Outlines:

- Introduction.
- Definition of breast examination.
- Purpose of breast examination.
- Indication of breast examination.
- Types of breast examination.
- Risk factors of breast illness.
- Equipment of breast examination.
- Procedure of breast examination (done by nurse -self-examination).
- Video on breast examination

Introduction

Breast examination is a vital clinical skill for early detection of breast cancer and other breast conditions. It involves both inspection and palpation techniques, performed by the patient (self-exam) or healthcare provider (clinical exam).

Breast cancer: 2nd most common cancer, 2nd leading cause of death, Incidence increases with age, all women are at risk of breast cancer.

Definition of breast examination:

A breast examination involves visual inspection and physical palpation of the breasts and surrounding areas to check for any abnormalities. This can be done by a healthcare professional (clinical breast exam) or by the individual (breast self-exam), to identify potential signs of breast cancer or other breast problems early on.

Purpose of breast examination

- To detect abnormalities of the breast such as mass, swelling or inflammation .and treat early abnormalities or complications.
- To assess breast size, shape, elasticity and symmetry
- To assess the nipple for it is type, size, and secretions.
- To assess the breast for signs of engorgement, mastitis or abscess.

Indications:

Breast examination is indicated in the following conditions:

1- At puberty at a regular basis for early detection of cancer breast.
2- In antenatal period for breast appraisal.
3- In postpartum period for detecting signs of engorgement or other breast complications in lactating mothers (after all milk has been expressed).
4- After menopause or following hysterectomy for early detection of cancer breast.
5- Breast examination indicated for women potentially develop cancer breast e.g. early menarche, over 40yearsofage, family history of cancer breast, nulliparous women whose first parity occurred after the age 34, exposure to carcinogens and presence of other cancer such as endometrial, colon, rectum, salivary gland and ovary.

Types of breast examination

1-Self-Breast Examination (SBE):

- Monthly check by the woman herself.
- Best done 7–10 days after menstruation.

2- Clinical Breast Examination (CBE):

- Performed by a trained healthcare professional.
- Part of routine physical exams.
- Recommended every 1–3 years (ages 25–39), annually after.

Risk Factors for Breast Illness

A-Uncontrollable Risk Factors:

- Gender.
- Age.
- Strong family history.
- History of previous breast biopsy which showed precancerous condition
- Nulliparity or first pregnancy after 30 years old
- Early menarche <10 years of age
- Late menopause >50 years of age
- Mutations of Breast cancer genes (BRCA 1 and BRCA 2).

B-Controllable Risk Factors:

- Diet, Alcoholic females.
- Currently or recently using combined estrogen and progesterone hormone replacement therapy.

Time of examination:

- After puberty don breast examination after the last days of period and at the same time each month (less tender, less congestion).
- BSE is best performed after menses (5th- 7th day) and first day of the month for postmenopausal women.

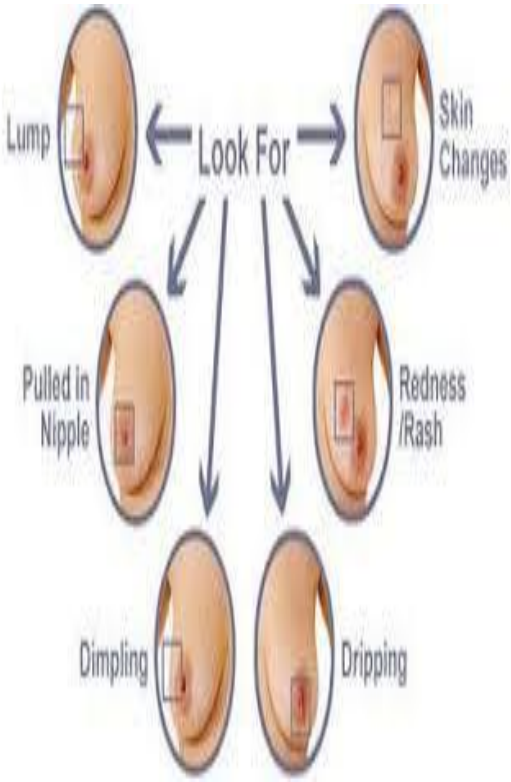
Equipment:

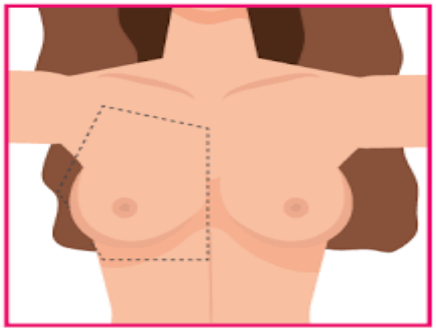
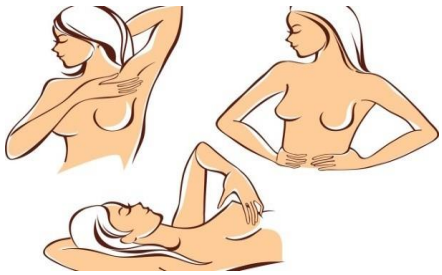
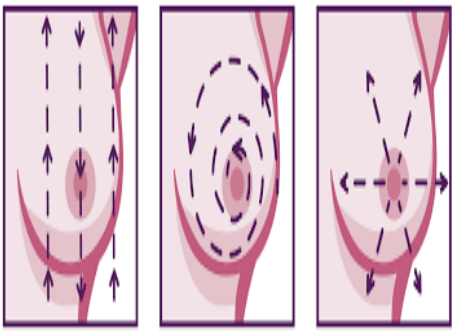
- Large Mirror, Good light lamp, Gloves, Small pillow or rolled towel and sterile gloves.

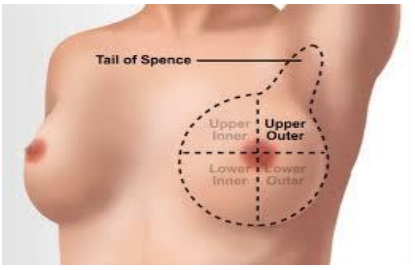
Breast Examination Guideline

STEP/TASK	Rational
Getting Ready 1. Prepare the necessary equipment and as- assemble it on the tray to bedside. 2. 2. Greet the women respectively. 3. 3. Explain procedure to the woman, what is going to examine her breasts, listen to her, and respond to her questions and concerns. 4. Discuss advantages of regular breast self- examination. 5. Ask client to empty bladder prior to procedure for comfort. 6. Close windows and keep privacy of the client to undress. 7. Raise the bed to the best level for good body mechanics. 8. Obtain detailed history from the client to identify possible risk factors for cancer breast regard: ✓ Age, menstrual history, gravid and parity. Development of the	To save time and effort.

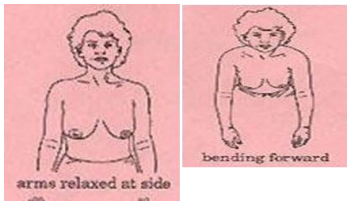
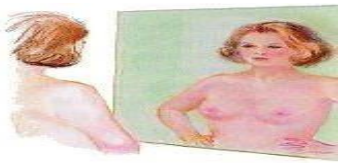
breast. ✓ Breast-feeding. ✓ Changes in size, shape or color of breast. Medications and any breast surgery. ✓ Soreness of breasts or nipples, particularly in relation to the menstrual cycle pain. ✓ Any discharge or secretions (describe color and consistency). ✓ Any lumps, masses, cysts or tumors (when noticed, how detected, how treated). ✓ Family history of breast cancer or breast disease. ✓ Any previous mammogram results. ✓ Knowledge of breast self-examination. ✓ Make sure good lightening. ✓ There should be no glares or shadows. 6. Wash hands and dry it then wear gloves	

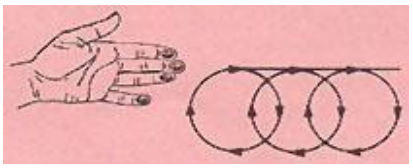
Breast Examination Done by the Nurse	
7. Ask the woman to undress from her waist up. 8. Sit her on the examining table with, her arms at her sides and facing the examiner to detect symmetry in size and observe color of skin, thickening or edema. 9. Inspect breast for size and symmetry. Note position of nipple in relation to underlying inter costal space observe area of chest over which breast extends. 10. Observe contour or shape of breast, noting any masses, retracting of tissue or flattening. If retraction of tissue suspected ask woman to raise arms above head or press hands against her hips. 11. Inspect overlying skin for color and venous pattern. If breasts are large, carefully lift breasts to inspect underlying skin surface. 12. Look at the nipples and note size, shape and direction in which they	

point. Check for rashes or sores and nipple discharge and milk or colostrum	
13. Repeat inspection in sitting position with arms raise over the head. This helps in contradiction of the pectoral muscles and does not alter the contour of the breast normally	
14. Examine the clavicle and axillary regions through palpation the tail of the breast and check for enlarged lymph nodes, tenderness, discoloration, swelling or lesions.	
15. Ask the client to assume the supine position and put her right arm over her head and palpate the breast tissue starting from outer edge of the upper quadrant at 12o'clock toward the nipple with using the circular or wedges or vertical techniques of the right breast. Note any lumps or tenderness. 	Normally lymph nodes are not palpate. Infection, inflammation and cancerous cell growth can cause lymph nodes to enlarge and become palpable. 

16. Repeat with the left breast and note any differences from the right breast.	To detect any abnormalites.
17. Make imaginary division of breast into 4 quarters.	
18. Use the palms of the hand: the flat sur- face or flat pads of the three middle fingers, rather than the finger tips to feel large areas of the breast against the chest wall because the finger tips tend to push a lump away. 19. Nodules caused by occluded milk ducts dur- ing puerperium can be palpated most fre quently in the upper outer quadrants	
20. if the mass is found palpate for following characteristics: location in relation to quad- rant size in centimeters; shape(e.g.) round or discoid; consistency (soft, firm or hard); tenderness; mobility	
21. If a lymph node is palpable; note	Characteristics of node help

its consistency, mobility, and presence of tenderness.	determine cause of enlargement such as cancerous nodes are hard, fixed, and non-tender. -while inflammation cause lymph Nodes enlarge and tender
22. Inspect and palpate the areola and nipples. 23. Gently spread the areola and squeeze the nipple between fingers to demonstrate the presence of cracks, fissures and secretion.	
24. Record normal and abnormal findings and date of next examination to refer the client 25. Physician if abnormalities are present.	
Breast Self-Examination Steps Motivate the client to perform her own breast examination regularly following the directions below because regular examination will show what is normal for the client and will give her confidence in	

Performing the procedure after menstrual period by (5-7) days as breasts during menses congested. 1-Stand in front of a mirror during showering 2-Inspect breast for size and symmetry. Note position of nipple in relation to underlying inter costal space observe area of chest over which breast extends.	
Inspect breasts with arms re laxed at your sides then with arms bending forward to detect any visual changes in size, shape, skin color or dimpling.	
4- Lift arms over head to look for any change in the con- tour of each breast, swilling, dimpling of skin or changes in the nipple.	
5- Rest the palm on hips and press down firmly to flex the chest muscles so that both breasts will hang evenly.	

6- Gently squeeze the nipple (Right and left) to look for any unusual discharge particularly pinkish or reddish discoloration.	
7-If you detect abnormal discharge, report immediately to doctor	
8-. Examine your breast during bath or shower by raising your left arm over your head to examine the left breast with your right and (the right breast with the left hand) because hands glide easily over wet skin.	
9- Be sure to use the flat pads of the three middle fingers rather than finger tips.	
10- Move gently and smoothly over very part of each breast using superficial and deep pressure of fingers with circular motion to check for any lump, hard knot, or thickening. To be sure you cover your whole breast, take your time and follow a definite pattern: **vertical**, **circles**, or **wedges**.	
11-after palpation of the breasts	

accurate locate the site of tumor or mass and its characteristics.	

Breast Examination Checklist

Action	First Trial		Second Trial	
	Done	Note Done	Done	Note Done
Getting Ready 1-Prepare the necessary equipment and as-assemble it on the tray to bedside, Large Mirror- Good light -Sterile gloves. 2-Greet the women respectively. 3- Explain procedure to the woman, what is going to examine her breasts, listen to her, and respond to her questions and concerns. 4- Discuss advantages of regular breast self- examination. 5- Ask client to empty bladder prior to procedure for comfort. 6- Close windows and keep privacy of the client to undress. 7- Raise the bed to the best level for good body mechanics. 8- Obtain detailed history from the client to identify possible risk factors for cancer breast regard: ✓ Age, menstrual history, gravid and parity. Development of the breast.				

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Score:				
Date:				

Student Name **Instructor signature**